

Instructions

Unless a question states otherwise, take $g = 9.8 \text{ m s}^{-2}$ wherever a numerical value of g is required. Solutions also give final answers for $g = 10 \text{ m s}^{-2}$ where these differ.

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1. (a) State the dimensions of force.

[1]

(b) The braking distance of a vehicle can be modelled by the formula

$$d = m^a u^b R^c$$

where,

m is the mass of the vehicle in kg

u is the initial speed in m s^{-1}

R is the constant braking force in N

Use dimensional analysis to find the values of a , b and c .

[4]

- 2.** A small satellite moves in a circular orbit of radius r about the centre of a much more massive planet of mass M . It is suggested that the period T of the orbit depends only on G , M and r , and that

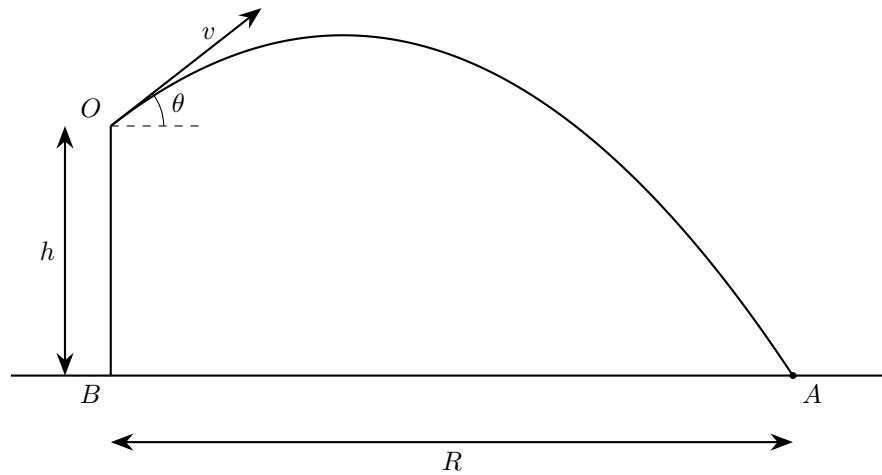
$$T = kG^\alpha M^\beta r^\gamma$$

where k is a dimensionless constant.

It is given that

$$G = 6.67 \times 10^{-11} \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2}$$

- (a) Write down the dimensions of G . [1]
- (b) By using dimensional analysis, determine the values of α , β and γ . [3]



3. A particle is projected with speed v from a point O at the top of a cliff of height h above horizontal ground. The angle of projection is θ above the horizontal. The particle lands at the point A on the ground, and B is the foot of the cliff, so that $BA = R$, as shown in the diagram.

You are given that

$$R = \frac{v^\alpha \cos \theta}{g^\beta} \left(v^\gamma \sin \theta + \sqrt{v^\delta \sin^2 \theta + 2gh} \right)$$

Use dimensional analysis to find α , β , γ and δ .

[4]

4. The increase in pressure p pascals between the surface and the bottom of a liquid of depth h metres and density $\rho \text{ kg m}^{-3}$ is modelled by

$$p = k\rho^\alpha h^\beta g$$

where $g \text{ m s}^{-2}$ is the acceleration due to gravity and k is a dimensionless constant.

- (a) State the dimensions of p . [1]
- (b) Use dimensional analysis to find the value of α and the value of β . [3]

5. The force, F , exerted on a flat plate by a water jet is thought to depend on the density of the water, ρ , the cross-sectional area of the jet, A , and the speed of the water, v . You are given that ρ is measured in kg m^{-3} .

(a) Find the dimensions of density ρ . [2]

A model of the force suggests the relationship

$$F = \frac{1}{2} \rho^\alpha A^\beta v^\gamma$$

(b) Explain whether or not it is possible to use dimensional analysis to verify that the constant $\frac{1}{2}$ is correct. [1]

(c) Use dimensional analysis to find the values of α , β and γ . [4]

6. A motor pulls a crate of mass m along a rough horizontal floor. The equation below is suggested to model the pulling force exerted by the motor, F , while the crate moves in a straight line.

$$F = k_1 m v \frac{dv}{dx} + k_2 \mu m g + k_3 \frac{Q}{x}$$

in which,

- v is the speed of the crate when it has travelled distance x
- μ is the coefficient of friction between the crate and the floor
- g is the acceleration due to gravity
- Q is the total thermal energy transferred to the surroundings after the crate has moved distance x
- k_1 , k_2 and k_3 are dimensionless constants

Determine whether the equation is dimensionally consistent.

[6]

7. The speed of a particle falling vertically through a resistive medium is modelled by the formula

$$v = \frac{mg}{k} \left(1 - e^{-kt/m} \right)$$

where,

v is the speed of the particle

m is the mass of the particle

g is the acceleration due to gravity

t is the time since the particle was released

k is a constant

Use dimensional analysis to determine the dimensions of k .

[4]

8. A comet of mass m measured in kg follows an elliptical orbit around a planet. The semi-major axis of the orbit is a measured in metres and the orbital period is T measured in seconds. The eccentricity e of the orbit is dimensionless.

A constant μ associated with the planet is defined by

$$T^2 = \frac{4\pi^2 a^3}{\mu}$$

and the angular momentum of the comet is given by

$$H = m\sqrt{\mu a(1 - e^2)}$$

By using dimensional analysis, find the dimensions of:

(a) μ

[3]

(b) H

[3]

9. A small ball of mass m kg is projected vertically upwards with speed u m s⁻¹. During the motion, when its speed is v m s⁻¹, it experiences an air resistance of magnitude kv^2 N, where k is a constant.

(a) Determine the dimensions of k .

[3]

The greatest height reached by the ball is H m. A formula approximating H is

$$H = \frac{u^2}{2g} - \frac{ku^4}{4mg^2} + \frac{1}{6}k^2m^\alpha u^\beta g^\gamma$$

where g m s⁻² is the acceleration due to gravity.

(b) Use dimensional analysis to find α , β and γ .

[4]

- 10.** A small probe is launched vertically upwards from the surface of a spherical moon. The radius of the moon is R metres, and the probe is at a distance r metres from the centre of the moon. The gravitational field strength g at distance r is modelled by

$$g = \frac{\Gamma}{r^2}$$

where Γ is a constant associated with the moon.

The probe has initial speed u at the surface and speed v when it reaches distance r . These speeds are related by

$$v^2 = u^2 - 2\Gamma \left(\frac{1}{R} - \frac{1}{r} \right)$$

The work done against gravity in moving a probe of mass m from the surface to distance r is

$$W = m\Gamma \left(\frac{1}{R} - \frac{1}{r} \right)$$

- (a) Find the dimensions of Γ . [2]
- (b) Use dimensional analysis to show that the equation for v^2 is dimensionally consistent. [2]
- (c) Find the dimensions and SI units of W . [2]